

Spotify Artist Streaming Analysis

Report type: Exploratory Data Analysis (EDA) **Dataset:** `dataset.csv` — 500 artists, 14 fields

1. Executive Summary

This report analyzes streaming performance for 500 recording artists across 23 genres and 43 countries, covering debut years from 1939 to 2023. The goal was to understand which genres, artist types, and eras drive the highest streaming numbers, and how collaboration behavior relates to performance.

Three headline findings stand out:

- Reggaeton, Pop, and Hip-Hop artists average the highest streams per artist**, despite having far fewer artists in the dataset than genres like Rock signalling concentrated, high-intensity fan engagement rather than broad volume.
 - Solo artists substantially out-earn groups in streams** solo artists average ~18,019M streams vs. ~13,290M for groups, a ~36% gap, even though the two distributions overlap heavily and both are right-skewed by superstar outliers.
 - Feature/collaboration streams are the strongest lever correlated with genre-level success** genres with the highest average collaboration rates (Filmi, EDM, Reggaeton) also cluster among the top genres by mean streams, and "Feature Streams" correlates more strongly with `Streams_Per_Year` (a proxy for career momentum) than raw genre alone.
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2. Dataset Overview

Attribute		Detail
Rows (artists)	500	
Columns	14	
Duplicate rows	0	
Duplicate artist names	0	
Missing values	0 across all columns	

Time span	Debut years 1939–2023
Genres	23 unique
Countries of origin	43 unique
Artist type split	Solo: 373 (74.6%) · Group: 127 (25.4%)
Sex split	Male: 390 · Female: 88 · Mixed (groups): 22
Total streams represented	~8.41 billion (millions-unit sum across all artists)
Mean streams/artist	~16,818M
Median streams/artist	~11,923M
Outlier artists (IQR method, Total Streams)	49

Note on mean vs. median: the mean is ~41% higher than the median, confirming a right-skewed distribution a small number of superstar artists pull the average well above what a "typical" artist in the dataset earns. This is why the report leans on median and rank-based comparisons alongside means.

Top 5 artists by total streams: Drake (137,492M), Taylor Swift (127,861M), Bad Bunny (125,900M), The Weeknd (95,703M), Justin Bieber (78,104M).

Geographic concentration: the dataset is heavily US/UK-weighted United States (265 artists, 53%) and United Kingdom (55 artists, 11%) together account for roughly two-thirds of all artists. This matters for interpreting genre and country findings (see Limitations, Section 6).

Data quality: the dataset has zero missing values and zero duplicate rows or artist names no cleaning or imputation was required beyond stripping whitespace from column headers.

3. Methodology

1. **Loading & cleaning** — data loaded via pandas; column names stripped of whitespace; null and duplicate checks confirmed a complete, clean dataset.
2. **Feature engineering** — added **Decade** (derived from Debut Year) and **Streams_Per_Year** (total streams normalized by career length, using 2026 as the reference year) to control for the fact that older artists have had more time to accumulate streams.
3. **Grouped aggregation** — streams summed/averaged by Genre, Sex, Artist Type, Country × Genre, and Decade.

4. **Distribution analysis** — compared Solo vs. Group streaming distributions using mean, median, and count, since raw streams are heavily right-skewed.
 5. **Outlier detection** — applied the IQR method (1.5× interquartile range) on Total Streams to flag statistically extreme performers separately from the general population.
 6. **Correlation analysis** — Pearson correlation matrix across all numeric fields (Debut Year, Total/Lead/Feature Streams, % Solo, % Collaborative, Streams Per Year) to identify which factors move together.
 7. **Time trend analysis** — debut counts by year to understand how the artist pool itself has grown, which contextualizes genre and decade comparisons.
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4. Key Findings

4.1 Genre performance

- **Reggaeton** leads all genres in mean streams per artist (~20,671M, among genres with at least 11 artists), ahead of **Pop** (~19,643M) and **Hip-Hop** (~18,487M).
- **R&B** and **EDM** round out the top five (~18,410M and ~16,686M respectively).
- Genres like **Latin**, **Regional Mexican**, and **Rock** sit meaningfully lower (~12,600–13,500M mean), despite Rock having one of the largest artist counts (60) in the dataset suggesting genre *intensity* (streams per artist) doesn't track genre *popularity by artist count*.
- By **total** streams (not averaged), **Hip-Hop** (~2.13B) and **Pop** (~2.00B) dominate simply because they have the most artists (115 and 102 respectively) this is a volume effect, distinct from the per-artist intensity story above.

4.2 Solo vs. Group

- Solo artists have both a higher median (~12,700M vs. ~10,607M) and higher mean (~18,019M vs. ~13,290M) than groups.
- Both distributions have long tails of high-performing outliers, but Solo's outliers reach further (up to ~137,492M) than Group's (topping out around ~45,000M) the ceiling for a breakout solo act is currently higher than for a breakout group in this dataset.
- The gap between medians is proportionally smaller than the gap between means, meaning artist type differentiates most strongly at the top end (superstar tier), not for the "typical" artist.

4.3 Debut year, streams, and career momentum

- There's **no strong linear relationship** between debut year and total streams (correlation coefficient: 0.07) an artist's era of debut doesn't by itself predict stream volume, since older artists have had longer to accumulate plays while newer artists benefit from a larger current listener base and algorithmic playlisting.

- **Artist debuts have grown sharply since 2000**, peaking around 2013–2017 (highs of 27–32 debuts/year) before a visible falloff post-2019 (likely a dataset recency/inclusion effect rather than a real-world decline see Limitations).
- **Streams_Per_Year correlates much more strongly with Total Streams (0.83) than Debut Year does (0.42)** meaning career-adjusted momentum is a far better indicator of an artist's absolute success than simply how long they've been active.

4.4 Correlation structure

- **Total Streams ↔ Lead Streams: 0.92** for most artists, lead/solo output is still the primary driver of total volume, even in collaboration-heavy genres.
- **Total Streams ↔ Streams_Per_Year: 0.83** career-adjusted momentum tracks closely with absolute success, i.e., high performers aren't just "high" because they've been active longer.
- **% Solo Streams ↔ % Collaborative Streams: -1.00** (by construction, since these are complementary percentages expected, not a discovery).
- **Debut Year ↔ % Collaborative Streams: 0.30** and **Debut Year ↔ Streams_Per_Year: 0.42** newer artists lean more collaborative and show higher per-year stream velocity, consistent with the industry's broader shift toward frequent features and playlist-driven collaboration since the 2010s.
- **Feature Streams ↔ % Collaborative Streams: 0.43** and **Feature Streams ↔ Total Streams: 0.65** feature-driven exposure meaningfully supports overall stream volume, though less than lead output does.

4.5 Collaboration by genre

- **Filmi (soundtrack/film music)** tops collaborative streaming share at ~96%, reflecting how that genre is structurally built around ensemble/soundtrack credits rather than solo releases.
- **EDM (~88%)** and **Reggaeton (~84%)** also skew heavily collaborative consistent with genre norms (producer + vocalist features, reggaeton's remix culture).
- Notably, **three of the four highest-collaboration genres (Reggaeton, EDM, Filmi) also appear in the top-streaming genres list (4.1)** collaboration intensity and streaming success move together at the genre level, even though at the individual-artist level, lead/solo output remains the stronger predictor (per 4.4).

4.6 Outliers

- 49 artists (~10% of the dataset) qualify as statistical outliers on Total Streams using the IQR method meaning roughly 1 in 10 artists sits far enough above the typical range to distort mean-based comparisons if not accounted for.
- The outlier list is topped by Drake, Taylor Swift, Bad Bunny, The Weeknd, Justin Bieber, Ariana Grande, Travis Scott, Kanye West, Ed Sheeran, and Eminem a mix of genres

(Hip-Hop, Pop, Reggaeton, R&B) confirming that extreme outliers aren't confined to a single genre.

5. Recommendations

These are framed as decision-relevant takeaways, useful either for a portfolio narrative or as if advising a label/artist strategy team:

1. **Prioritize strategic features over full pivots to group format.** Since feature streams correlate positively with total success (0.65) but lead streams correlate more strongly (0.92), the data supports artists supplementing not replacing a strong solo catalog with collaborations, rather than assuming group-based output outperforms.
 2. **Treat genre selection/positioning as a momentum signal, not a volume guarantee.** Reggaeton, EDM, and Filmi show high per-artist streaming intensity *and* high collaboration norms genres/scenes with strong feature culture appear to reward participants more per capita, even with fewer total artists competing for attention.
 3. **Use `Streams_Per_Year` (or an equivalent time-normalized metric), not raw totals, when comparing artists across different career lengths.** Raw Total Streams conflates "career length" with "success," and this dataset shows debut year alone barely explains total streams ($r = 0.07$), while career-adjusted momentum explains far more ($r = 0.83$).
 4. **Don't over-index on the "genre by artist count" story.** High-artist-count genres (Rock, Pop) aren't automatically the highest per-artist earners Reggaeton beats them despite a smaller artist base. Any resourcing or A&R strategy based on genre popularity should separate "how many artists are in this genre" from "how well does the average artist in this genre perform."
 5. **Model or report on outliers separately from the typical artist.** With ~10% of the dataset qualifying as statistical outliers, any predictive modeling (e.g., predicting a new artist's expected streams) should either use robust/median-based methods or explicitly segment superstar-tier artists from the broader population, so headline averages aren't quietly driven by a handful of names.
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6. Limitations & Caveats

- **Sampling/selection bias:** the dataset is dominated by US (53%) and UK (11%) artists; genre and country conclusions should be read as "within this artist sample," not as a representative view of the global streaming market.
- **Survivorship bias:** artists included are, by definition, ones with enough streams to be tracked/notable the dataset says nothing about failed or low-visibility artists in each

genre, so "genre X performs well" really means "successful artists tagged genre X perform well."

- **Post-2019 debut drop-off is very likely a data recency/inclusion artifact**, not a real decline in new artist debuts newer artists haven't had time to accumulate comparable streaming totals, and datasets like this typically undersample very recent, still-emerging acts.
 - **Correlation $\neq$ causation**: e.g., the collaboration–genre streaming relationship (4.5) is an association across genre-level averages, not evidence that adding features *causes* higher streams for an individual artist.
 - **No statistical significance testing** was performed (e.g., no t-test between Solo vs. Group medians). The report describes observed differences in the sample, not statistically validated population effects.
 - **"% of Solo/Collaborative Streams" are complementary by construction** (they sum to 100%), so their -1.00 correlation is a structural fact of the data, not a discovered relationship this is called out above to avoid over-interpreting it as a finding.
 - **Static snapshot**: streaming totals are cumulative as of the data pull, not a time series, so nothing here reflects trends *within* an artist's career (e.g., recent momentum vs. lifetime average).
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7. Suggested Next Steps

- Add a regression or tree-based model (e.g., predicting **Total Streams** or **Streams_Per_Year** from Genre, Artist Type, Country, and Collaboration %) to move from descriptive to predictive analysis.
- Run a formal significance test (t-test or Mann-Whitney U, given the skew) on the Solo vs. Group difference to confirm it's not just sampling noise.
- Re-run genre/type comparisons excluding the 49 identified outliers to see if conclusions hold for "typical" artists.
- If available, bring in a per-track or per-release dataset (rather than artist-level totals) to analyze release strategy (frequency, feature rate) instead of just cumulative outcomes.
- Expand geographic coverage or explicitly weight/report findings by region, given the current US/UK concentration.